

HCI Notes: Designs and Evaluations

HCI - Week 5

Abstract

These notes cover the intertwined processes of design and evaluation in Human-Computer Interaction (HCI). The document is divided into three main parts. Part I outlines the key stages of the design process, from conceptual to physical design, while warning against technology myopia. Part II delves into evaluation, discussing the evolution from traditional to modern metrics, the importance of triangulation in research, and an overview of various data collection and analysis methods, including experimental research, surveys, and ethnographic approaches. Part III provides a comprehensive guide to design rules, distinguishing between principles, guidelines, and standards, and offers a detailed examination of two seminal sets of usability heuristics: Shneiderman's 8 Golden Rules and Nielsen's 10 Usability Heuristics.

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Part I

The Design Process

1 Overview of Design Stages (Slides 2-4)

The design of an interactive system is a multifaceted process that involves several stages, from high-level conceptualization to concrete physical implementation. A successful process also requires constant evaluation and adherence to established principles.

1.1 Technology Myopia

Warning: Technology Myopia

Technology Myopia is the flawed approach of focusing too narrowly on the technological aspects of a system while ignoring the human and social context in which it will be used.

To avoid this, interaction designs must be sensitive to:

- **Human-human communication:** Technology often mediates communication between people. Its design must support, not hinder, these social interactions.
- **Implicit Knowledge:** Much of human expertise and workflow is based on tacit or implicit knowledge that is hard to articulate. A good design respects and integrates these unspoken practices.
- **Non-technical aspects of work:** The real-world environment is messy and involves emotions, politics, and workarounds. A purely technical solution that ignores this reality is likely to fail.

1.2 Conceptual Design

This is the initial, high-level phase of design. It is concerned with defining the fundamental concepts and structure of the system.

- **Structuring the information space:** Organizing the data and functions in a logical and understandable way.
- **Creating alternative solutions:** Brainstorming and exploring multiple different design concepts to solve the core problem.
- **Determining which concept to pursue:** Evaluating the alternatives against user needs and project goals to select the most promising direction.

1.3 Physical Design

This phase translates the chosen conceptual design into a concrete, tangible form. It moves from the abstract "what" to the specific "how." Physical design addresses:

- **What it will look like:** Defining the visual aesthetics, branding, color palette, and typography.
- **What components it will require:** Selecting specific UI elements like buttons, menus, forms, and navigation bars.
- **How the screens will be laid out:** Creating the detailed screen-by-screen layout and workflow.

1.4 Standards and Tools

Physical design is greatly aided by the use of standards and development tools, which ensure consistency and efficiency.

- **Graphical libraries and UI toolkits:** Pre-built sets of common UI components (e.g., Material Design, Apple's Human Interface Guidelines) that enforce a consistent look and feel.

- **Visual interface builders and Web development tools:** Software that allows designers and developers to construct the interface visually and often generates the underlying code.

Part II

Evaluation and Measurement

2 Measuring the User Experience (Slides 5-6)

Evaluation is the process of assessing a design to ensure it meets user needs. This requires measurement, but what we measure has evolved over time.

2.1 Traditional Measurements

The classic trifecta of usability measurement includes:

1. **Task performance (Effectiveness):** Can users successfully complete their tasks? (e.g., percentage of tasks completed).
2. **Time performance (Efficiency):** How long does it take users to complete tasks?
3. **User satisfaction:** How do users feel about using the system? (Often measured with questionnaires like the System Usability Scale - SUS).

2.2 Limitations and New Forms of Measurement

These traditional metrics, while useful, fail to capture the full complexity of the modern user experience. They do not accurately measure:

- **Why** people abandon an interface.
- **Discretionary use for enjoyment** (e.g., watching YouTube, playing a game).
- Complex human factors like **emotion and trust**.

Therefore, new forms of measurement are needed to evaluate aspects like:

- A feeling of **community**.
- **Enjoyment** and emotional engagement.
- **Satisfaction** from accomplishments in virtual worlds.
- **Physiological measures** (e.g., EEG for cognitive load, EMG for muscle tension) to get objective data on user state.

3 Research Principles and Methods (Slides 7-13)

3.1 Triangulation

The Principle of Triangulation

No single data collection method is perfect. **Triangulation** is the research practice of using multiple methods and/or multiple researchers to investigate the same phenomenon.

The goal is to see if different approaches lead to the same conclusion. When they do, it builds a strong **consensus** and increases confidence in the findings. A single study is not "scientific truth"; robust conclusions come from the convergence of evidence from multiple sources.

3.2 Types of Research

There are three primary types of research, each with its own focus and typical methods.

Type of Research	Focus	General Claim	Typical Methods
Descriptive	Describe a situation or set of events.	"X is happening"	Observations, field studies, focus groups, interviews.
Relational	Identify relations between multiple variables.	"X is related to Y"	Observations, field studies, surveys.
Experimental	Identify causes of a situation or a set of events.	"X is responsible for Y"	Controlled experiments.

3.3 Experimental Design

Experimental design is the gold standard for determining cause-and-effect relationships. The type of experiment depends on key factors:

- **Is random assignment used?** If yes, it's a **true experiment**. If no, it's not. Random assignment is crucial for eliminating confounding variables.
- **Are there multiple groups or measures?** If a study lacks random assignment but still compares groups or takes multiple measurements over time, it is a **quasi-experiment**. If it has neither, it is a **non-experiment** (e.g., a simple one-off usability test).

When designing an experiment, one must also define the **independent variables** (the factors being manipulated) and their different **values** or levels.

3.4 Other Data Collection Methods

- **Surveys:** Excellent for getting a large number of responses quickly from a geographically dispersed population. They are good for capturing the "big picture."
- **Diaries (Ethnography):** Useful for recording information that is fluid and changes over time (e.g., user mood, technology use in daily life). This method is ideal when after-the-fact recall would be unreliable.
- **Case Studies:** An in-depth study of a specific instance within a real-life context. While time-intensive, they are valuable for building deep understanding and generating hypotheses.
- **Interviews and Focus Groups:** Directly asking users about their experiences, opinions, and feelings.
- **Ethnography:** A holistic method involving observation and participation within a user's natural environment to gain a deep understanding of their culture, behaviors, and needs.

Part III

Design Rules and Heuristics

4 Types of Design Rules (Slides 14-17)

The ultimate goal of interaction design is to achieve maximum **usability**. Design rules provide guidance to help achieve this goal. They can be classified along two dimensions:

- **Authority:** How strictly the rule must be followed. High authority rules are mandatory, while low authority rules are suggestions.
- **Generality:** How broadly the rule can be applied. High generality rules apply to many situations, while low generality rules are for specific contexts.

This gives rise to a spectrum of design rules:

1. **Standards:** High authority, low generality. These are specific, mandatory rules for a particular platform or organization (e.g., an ISO standard, a company's strict design system).
2. **Guidelines:** Lower authority, high generality. These are more general recommendations of good practice that are not strictly mandatory (e.g., "Provide informative feedback").
3. **Principles:** Lower authority, high generality. These are abstract, fundamental truths about design that can be interpreted in various ways (e.g., "Support the user's mental model").

5 Golden Rules and Heuristics (Slides 18-32)

Heuristics are "broad brush" design rules that act as a useful checklist for good design. They are not specific rules but general principles that have been proven effective over time.

5.1 Shneiderman's 8 Golden Rules

Developed by Ben Shneiderman, these rules are a highly influential set of design principles.

1. **Strive for Consistency:** Interfaces should be consistent both visually (color, fonts, layout) and behaviorally (similar actions should work the same way across the application). Consistency reduces cognitive load and makes an interface predictable and easy to learn.
2. **Cater to Universal Usability:** Designs should cater to a wide range of users, from novices to experts. This often means providing shortcuts (e.g., keyboard shortcuts, macros) for frequent users to increase their efficiency, while still keeping the interface simple for beginners.
3. **Offer Informative Feedback:** For every user action, there should be a clear and timely system reaction. This tells the user that their action was received and what its result was. Feedback can be a simple visual change, a message, or a sound.
4. **Design Dialogs to Yield Closure:** Sequences of actions should have a clear beginning, middle, and end. The user should receive a clear signal that they have completed a task (e.g., a "Your message has been sent" confirmation).
5. **Offer Error Prevention and Simple Error Handling:** The best design prevents errors from happening in the first place (e.g., by graying out unavailable options). If an error does occur, the system should present a clear, constructive message that explains the problem and helps the user recover.
6. **Permit Easy Reversal of Actions:** Users should be able to undo and redo their actions. This reduces anxiety and encourages exploration, as users know they can easily backtrack from a mistake.
7. **Support Internal Locus of Control:** Users should feel that they are in control of the system, not the other way around. The system's behavior should be predictable, and the user should be the initiator of actions.
8. **Reduce Short-Term Memory Load:** Human short-term (working) memory is limited. Interfaces should not require users to remember information from one screen to the next. Design for **recognition over recall** by making information and options visible. Information should be "chunked" into smaller, manageable groups to be easier to process.

5.2 Nielsen's 10 Usability Heuristics

Developed by Jakob Nielsen, these are perhaps the most widely used set of usability heuristics in the industry. They overlap with Shneiderman's rules but have a slightly different focus.

1. **Visibility of system status:** The system should always keep users informed about what is going on, through appropriate feedback within a reasonable time. (*Example: A progress bar during a file download.*)

2. **Match between system and the real world:** The system should speak the user's language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order. (*Example: A shopping cart icon for an e-commerce site.*)
3. **User control and freedom:** Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo. (*Example: A "Cancel" button on a multi-step form.*)
4. **Consistency and standards:** Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions. (*Example: Using a standard icon for "settings" across all screens.*)
5. **Error prevention:** Even better than good error messages is a careful design which prevents a problem from occurring in the first place. (*Example: Disabling the "Submit" button until all required form fields are filled.*)
6. **Recognition rather than recall:** Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. (*Example: Showing a list of recently viewed items.*)
7. **Flexibility and efficiency of use:** Accelerators — unseen by the novice user — may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to customize frequent actions. (*Example: Keyboard shortcuts like Ctrl+S to save.*)
8. **Aesthetic and minimalist design:** Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility. (*Example: A clean, uncluttered homepage with a clear call to action.*)
9. **Help users recognize, diagnose, and recover from errors:** Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution. (*Example: Instead of "Error -27B," show "Invalid email address. Please include an '@' symbol."*)
10. **Help and documentation:** Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large. (*Example: Contextual help icons next to complex form fields.*)